

Madee Uththara Deegoda Gamage

Data Analytics and AI Engineer | Machine Learning, Deep Learning, NLP

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PROFILE

Recent ICT graduate specialized in Data Analytics and AI, seeking junior roles in Data Science, Data Engineering, Machine Learning, and AI. Hands-on experience building and deploying end-to-end ML systems, from data pipelines and model training to FastAPI services and live demos. Bachelor's thesis on multilingual NLP, delivered as a production service for a university strategic project. Comfortable across the full stack when needed, with a strong mathematics and engineering foundation.

EDUCATION

Bachelor of Engineering, Information and Communications Technology

2023 - 2026

JAMK University of Applied Sciences, Finland | Specialization: Data Analytics and AI

- Core focus: Machine Learning, Deep Learning, Data Preprocessing, Data Analysis and Visualization, Computational Algorithms, all graded 5/5.
- Strong mathematics and engineering base: Discrete Mathematics, Vectors and Matrices, Optimization and Network Models, Probability and Statistics, Cryptology.
- Thesis: Evaluating Multilingual Sentence Embedding Models for Bilingual Course Description Similarity in Finnish Higher Education.

EXPERIENCE

Machine Learning Trainee

Feb 2026 - May 2026

JAMK University of Applied Sciences, LuoVuutta project, Institute of Information Technology

- Designed and implemented an end-to-end Python evaluation pipeline for multilingual sentence embedding models in a Finnish-English cross-lingual NLP task (course description similarity).
- Built a bilingual evaluation dataset from JAMK's Peppi curriculum system, benchmarked multiple multilingual embedding models and configurations, and applied cosine similarity scoring, cross-validation, threshold calibration, confidence intervals, AUC analysis, and statistical testing, including robustness and generalisation analyses.
- Researched and prototyped a FastAPI microservice serving the LaBSE model to integrate ML capabilities into the AI Curriculum Tool.
- **Tools:** Python, PyTorch, Hugging Face Transformers, Sentence-Transformers, scikit-learn, spaCy, pandas, NumPy, Jupyter, Git, Docker.

BACHELOR'S THESIS

Multilingual Sentence Embedding Evaluation for Bilingual Course Similarity

2026

Grade 5/5. Conducted within the LuoVuutta project (see Experience above).

- Benchmarked 5 sentence embedding models (LaBSE, STSB, E5, MPNet, FinBERT) across 30 configurations. LaBSE achieved the best results (cross-validated F1 0.976, AUC 0.992) and generalised to an independent held-out programme without fine-tuning.
- Live demo deployed on Hugging Face Spaces: [course-similarity-demo](#)

SELECTED PROJECTS

AI vs Human Text Detection (Deep Learning)

2025

- Compared 5 deep learning architectures (BiLSTM, 1D-CNN, Gated Convolution Unit, Hybrid, custom Transformer) for AI-generated text detection under identical training conditions.
- Best model (custom Transformer) reached 98.4% accuracy. Built with TensorFlow and Keras.

Used Car Price Prediction (End-to-End ML)

2025

Group project. Led business understanding, modeling, and deployment.

- Full CRISP-DM pipeline on 3 million listings using Polars. Compared 8 models; tuned Random Forest on log price reached R2 0.9456 (MAE approx 2,442 USD).
- Deployed the final model as a FastAPI REST API with a Streamlit interface, validated on 20,000 real records.

CV Screener (LLM Application)

2026

- AI-powered CV-to-job match scorer using Llama 3.1 via Groq, with structured JSON output, PDF parsing, and a Streamlit interface.

Finland Labour Market Dashboard (Data Visualization)

2026

- Interactive multi-tab dashboard on Finnish employment and housing data from Statistics Finland, OECD, and Eurostat. Live: uththara15.github.io/finland-employment-data

Expense Tracker (Full Stack)

2025

- Secure REST API with Node.js, Express, PostgreSQL, JWT authentication, node-cron recurring jobs, and external currency API integration. React and Vite frontend, deployed on Render.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, HTML/CSS
- **ML / AI:** Machine Learning, Deep Learning, NLP, sentence-transformers, PyTorch, TensorFlow, Keras, Scikit-learn, spaCy, LLM integration (Groq)
- **Data:** Pandas, NumPy, Polars, DuckDB, Power BI, Plotly, Chart.js
- **Deployment / Tools:** FastAPI, Flask, Streamlit, Docker, Hugging Face Spaces, Git, Node.js, Express, PostgreSQL, React

LANGUAGES

- English: fluent (entire degree taught in English; working language during traineeship was English).
- Finnish: basic (Finnish 1 and 2 completed). Sinhala: native.
- Completed Talent Boost: Integration into Finnish Society and Creating Careers programmes.